

Mathematical model: hybrid CSP + PVT plant sizing

Assignment 1, Question 5

1. Sets

Symbol	Definition
Days	{June, December}
Hrs	{0, 1, ..., 23}
SunHrs _{day}	{hr ∈ Hrs : Irradiance _{hr,day} > 0}, the sunlit hours of each day

2. Parameters

Data, for all hr ∈ Hrs and day ∈ Days:

Symbol	Meaning	Unit
Irradiance _{hr,day}	solar irradiance	W/m ²
T _{ambient,hr,day}	ambient air temperature	°C
Demand _{elec,hr,day}	electricity demand in that hour	GJ
Demand _{heat,hr,day}	heat demand in that hour	GJ

CSP

Symbol	Meaning	Value
Eff _{reflectivity}	heliostat mirror reflectivity	0.95
Eff _{cosine}	average cosine efficiency	0.86
Eff _{shading blocking}	shading and blocking efficiency	0.89
Eff _{atmospheric}	atmospheric attenuation efficiency	0.97
Eff _{spillage}	spillage efficiency	0.92
Eff _{receiver}	receiver absorption efficiency	0.80
Eff _{rankine}	Rankine cycle thermal efficiency	0.42
Area _{per heliostat}	mirror area of one heliostat	220 m ²
Cost _{CSP}	heliostat cost per m ²	€80/m ²

PVT

Symbol	Meaning	Value
$\text{Eff}_{\text{PVT,elec}}$	PVT electrical efficiency	0.16
$\text{Frac}_{\text{absorbed}}$	effective optical fraction	0.86
$\text{Coef}_{\text{heat loss}}$	overall heat loss coefficient	12 W/(m ² ·K)
$\text{Factor}_{\text{heat removal}}$	collector heat removal factor	0.65
$T_{\text{water in}}$	inlet water temperature	35 °C
$\text{FlowRate}_{\text{per PVT panel}}$	water mass flow rate through one panel	0.2 kg/s
$c_{\text{p,water}}$	specific heat of water (standard value)	4186 J/(kg·K)
$\text{Area}_{\text{per PVT panel}}$	area of one PVT panel	4 m ²
Cost_{PVT}	PVT cost per m ²	€150/m ²

Plant

Symbol	Meaning	Value
Area_{max}	maximum available area	450 000 m ²
$\text{Share}_{\text{CSP,min}}$	minimum CSP share of electricity	0.60
$\text{GJ}_{\text{per W-hr}}$	energy of 1 W sustained for 1 hour	3.6×10^{-6} GJ

3. Derived parameters

Efficiency chains

$$\begin{aligned}\text{Eff}_{\text{optical}} &= \text{Eff}_{\text{reflectivity}} \cdot \text{Eff}_{\text{cosine}} \cdot \text{Eff}_{\text{shading blocking}} \cdot \text{Eff}_{\text{atmospheric}} \cdot \text{Eff}_{\text{spillage}} \\ \text{Eff}_{\text{CSP}} &= \text{Eff}_{\text{optical}} \cdot \text{Eff}_{\text{receiver}} \cdot \text{Eff}_{\text{rankine}}\end{aligned}$$

Yield per m², for all $\text{hr} \in \text{Hrs}$ and $\text{day} \in \text{Days}$ [W/m²]

$$\begin{aligned}\text{Yield}_{\text{CSP elec,hr,day}} &= \text{Eff}_{\text{CSP}} \cdot \text{Irradiance}_{\text{hr,day}} \\ \text{Yield}_{\text{PVT elec,hr,day}} &= \text{Eff}_{\text{PVT,elec}} \cdot \text{Irradiance}_{\text{hr,day}} \\ \text{Yield}_{\text{PVT heat,hr,day}} &= \max\left(0, \text{Factor}_{\text{heat removal}} \left[\text{Frac}_{\text{absorbed}} \cdot \text{Irradiance}_{\text{hr,day}} \right. \right. \\ &\quad \left. \left. - \text{Coef}_{\text{heat loss}} (T_{\text{water in}} - T_{\text{ambient,hr,day}}) \right] \right)\end{aligned}$$

This is the heat-removal-factor form of the flat plate collector, $\dot{Q}_u = F_R A_c [(\tau\alpha)_{\text{eff}} I_T - U(T_{\text{in}} - T_a)]$ with $A_r = A_c$, divided by the collector area. Losses are evaluated at the inlet water temperature; $\text{Factor}_{\text{heat removal}}$ corrects for the plate being warmer.

PVT water outlet temperature, for all $\text{hr} \in \text{Hrs}$ and $\text{day} \in \text{Days}$ [°C]

$$T_{\text{water out,hr,day}} = T_{\text{water in}} + \frac{\text{Area}_{\text{per PVT panel}} \cdot \text{Yield}_{\text{PVT heat,hr,day}}}{\text{FlowRate}_{\text{per PVT panel}} \cdot c_{\text{p,water}}}$$

4. Decision variables

Symbol	Meaning	Domain
$\text{Count}_{\text{heliostats}}$	number of heliostats	$\{0, 1, 2, \dots\}$
$\text{Count}_{\text{PVT panels}}$	number of PVT panels	$\{0, 1, 2, \dots\}$

with the resulting areas

$$\text{Area}_{\text{CSP}} = \text{Area}_{\text{per heliostat}} \cdot \text{Count}_{\text{heliostats}} \quad \text{Area}_{\text{PVT}} = \text{Area}_{\text{per PVT panel}} \cdot \text{Count}_{\text{PVT panels}}$$

5. Full model

Supply, for all $\text{hr} \in \text{Hrs}$ and $\text{day} \in \text{Days}$ [GJ]

$$\begin{aligned} \text{Supply}_{\text{CSP elec,hr,day}} &= \text{GJ}_{\text{per W-hr}} \cdot \text{Area}_{\text{CSP}} \cdot \text{Yield}_{\text{CSP elec,hr,day}} \\ \text{Supply}_{\text{PVT elec,hr,day}} &= \text{GJ}_{\text{per W-hr}} \cdot \text{Area}_{\text{PVT}} \cdot \text{Yield}_{\text{PVT elec,hr,day}} \\ \text{Supply}_{\text{PVT heat,hr,day}} &= \text{GJ}_{\text{per W-hr}} \cdot \text{Area}_{\text{PVT}} \cdot \text{Yield}_{\text{PVT heat,hr,day}} \end{aligned}$$

Objective function

$$\min \text{Cost}_{\text{total}} = \text{Cost}_{\text{CSP}} \cdot \text{Area}_{\text{CSP}} + \text{Cost}_{\text{PVT}} \cdot \text{Area}_{\text{PVT}}$$

Constraints

Heat demand, for all $\text{day} \in \text{Days}$:

$$\sum_{\text{hr} \in \text{SunHrs}_{\text{day}}} \text{Supply}_{\text{PVT heat,hr,day}} \geq \sum_{\text{hr} \in \text{SunHrs}_{\text{day}}} \text{Demand}_{\text{heat,hr,day}} \quad (\text{C1})$$

Electricity demand, for all $\text{day} \in \text{Days}$:

$$\sum_{\text{hr} \in \text{SunHrs}_{\text{day}}} (\text{Supply}_{\text{CSP elec,hr,day}} + \text{Supply}_{\text{PVT elec,hr,day}}) \geq \sum_{\text{hr} \in \text{SunHrs}_{\text{day}}} \text{Demand}_{\text{elec,hr,day}} \quad (\text{C2})$$

Minimum CSP share, for all $\text{day} \in \text{Days}$:

$$\begin{aligned} \sum_{\text{hr} \in \text{SunHrs}_{\text{day}}} \text{Supply}_{\text{CSP elec,hr,day}} \\ \geq \text{Share}_{\text{CSP,min}} \sum_{\text{hr} \in \text{SunHrs}_{\text{day}}} (\text{Supply}_{\text{CSP elec,hr,day}} + \text{Supply}_{\text{PVT elec,hr,day}}) \end{aligned} \quad (\text{C3})$$

Available area:

$$\text{Area}_{\text{CSP}} + \text{Area}_{\text{PVT}} \leq \text{Area}_{\text{max}} \quad (\text{C4})$$

Integrality and non-negativity:

$$\text{Count}_{\text{heliostats}}, \text{Count}_{\text{PVT panels}} \in \{0, 1, 2, \dots\} \quad (\text{C5})$$